

Technical Data Sheet

BondMax 500

High-Performance Industrial Adhesive

Product Code: BM-500 Revision: 4.2 Date: October 2025

Product Description

BondMax 500 is a two-component, high-strength structural adhesive designed for demanding industrial bonding applications. It provides exceptional bond strength on metals, plastics, and composites.

Applications

- Metal-to-metal bonding
- Automotive assembly
- Industrial equipment manufacturing
- Structural repairs
- Heavy machinery assembly

Technical Properties

Uncured Properties

Property	Value	Test Method
Base Color	Light amber	Visual
Hardener Color	Clear	Visual
Mix Ratio	2:1 (Base:Hardener)	By volume
Viscosity (mixed)	2,500-3,500 cP	ASTM D2196
Density	1.05-1.10 g/cm ³	ASTM D1475
Pot Life (100g)	45 minutes	25°C
Application Temp	15-35°C	-

Cured Properties (at 25°C)

Property	Value	Test Method
Cure Time (handling)	4-6 hours	-
Full Cure	24 hours	-
Tensile Strength	> 15 MPa	ASTM D638
Shear Strength (steel)	> 12 MPa	ASTM D1002
Peel Strength	> 8 N/mm	ASTM D1876
Shore D Hardness	75-85	ASTM D2240
Service Temp Range	-40°C to +120°C	-

Surface Preparation

1. Remove all dirt, oil, grease, and loose particles
2. Abrade smooth surfaces with 80-120 grit sandpaper
3. Clean with isopropyl alcohol or acetone
4. Apply adhesive within 1 hour of preparation

Application Instructions

1. Ensure surfaces are clean and dry
2. Dispense base and hardener at 2:1 ratio
3. Mix thoroughly for 2-3 minutes
4. Apply to one or both surfaces
5. Assemble parts and clamp if necessary
6. Allow minimum 4 hours before handling
7. Full cure achieved after 24 hours at 25°C

Storage & Shelf Life

Condition	Requirement
Storage Temperature	15-25°C
Humidity	< 60% RH
Shelf Life (unopened)	12 months
Shelf Life (opened)	6 months

Packaging

Size	Product Code
25 kg drum	BM-500-25
200 kg barrel	BM-500-200
400 ml cartridge	BM-500-CART

Safety Information

- Use in well-ventilated areas
 - Wear protective gloves and safety glasses
 - Avoid skin contact
 - See Safety Data Sheet for complete information
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